

A Short Guide to Appendix B

Young local limits and randomized rounding

Abstract

Appendix B proves that the Schur–Weyl Young-label law has a uniform Gaussian local limit and that randomized dilation of a label produces the correctly dilated Gaussian limit. The first lemma compares the Young law with a multinomial law and then proves a cellwise local central limit theorem. The second lemma shows that the randomized rounding kernel becomes ordinary Gaussian dilation in central-limit coordinates. A final typical-set argument proves that the fallback correction is asymptotically irrelevant and that all typical input–output labels satisfy the required branching condition. This note explains the geometry of the lattice cells, both lemmas, and the final proposition.

1 Purpose of Appendix B

Let

$$\Delta_d^\circ = \left\{ p \in \mathbb{R}^d : p_1 > \cdots > p_d > 0, \sum_i p_i = 1 \right\},$$

and set $\rho_p = \text{diag}(p_1, \dots, p_d)$. Under the Schur–Weyl decomposition

$$(\mathbb{C}^d)^{\otimes N} \cong \bigoplus_{\lambda \in \mathbf{Y}_N} \mathcal{H}_\lambda \otimes \mathcal{S}_\lambda,$$

let Π_λ denote the projection onto the summand indexed by λ . The Young-label probability is

$$P_{N,p}(\lambda) := \Pr_{\rho_p^{\otimes N}} \{\text{label } \lambda\} = \text{Tr}(\Pi_\lambda \rho_p^{\otimes N}) = \dim \mathcal{S}_\lambda s_\lambda(p), \quad \lambda \in \mathbf{Y}_N.$$

Here $\mathbf{Y}_N$ is the set of partitions of N with at most d rows, $\mathcal{H}_\lambda$ is the corresponding irreducible $U(d)$ module, $\mathcal{S}_\lambda$ is the Specht module, and $s_\lambda(p) = \text{Tr}_{\mathcal{H}_\lambda} \pi_\lambda(\rho_p)$ is the Schur polynomial evaluated at p . More precisely, if $\lambda \sim P_{N,p}$, then $(\lambda - Np)/\sqrt{N}$ converges to the centered Gaussian $\mathbf{N}_{0,\Sigma_p}$ on $\mathbf{H}$, where $\Sigma_p = (\text{diag}(p) - pp^\top)|_{\mathbf{H}}$; Lemma B.1 strengthens this to uniform L^1 convergence after cell embedding. With respect to the induced Lebesgue measure on $\mathbf{H}$,

$$\mathbf{N}_{0,\Sigma_p}(\mathrm{d}x) = \varphi_{0,\Sigma_p}(x) \mathrm{d}x = \frac{\exp\left[-\frac{1}{2}\langle x, \Sigma_p^{-1}x \rangle\right]}{(2\pi)^{(d-1)/2} \sqrt{\det_{\mathbf{H}} \Sigma_p}} \mathrm{d}x.$$

For unknown spectrum, the channel cannot use p to sample a fresh target label. Instead, starting from an observed input label $\mu \in \mathbf{Y}_n$, it dilates μ by

$$\gamma_n = \frac{m_n}{n} \longrightarrow \gamma > 1$$

and randomly rounds the result to the target lattice. Appendix B proves two facts needed for this construction:

1. $P_{N,p}$ has the same uniform local Gaussian limit as the multinomial law with parameter p ;
2. randomized rounding sends this limit to the Gaussian with covariance $\gamma\Sigma_p$ and has asymptotic affinity $B_{\text{cl}}(\gamma, d)$ with the target law $P_{m_n,p}$.

The last part of the appendix verifies that, on the typical set, the rounded label is a valid partition and satisfies the representation-theoretic branching condition.

2 The affine lattice and its fundamental cell

Put

$$\mathbf{H} = \left\{ x \in \mathbb{R}^d : \sum_i x_i = 0 \right\}, \quad \mathbf{L}_N = \left\{ x \in \mathbb{Z}^d : \sum_i x_i = N \right\}.$$

The lattice parallel to $\mathbf{H}$ is generated by $b_i = e_i - e_d$, $1 \leq i < d$. Appendix B uses the half-open fundamental cell

$$\mathbf{C} = \left\{ \sum_{i=1}^{d-1} t_i b_i : -\frac{1}{2} \leq t_i < \frac{1}{2} \right\} \subset \mathbf{H}. \quad (1)$$

Write $\mathbf{A}_N = \{x \in \mathbb{R}^d : \sum_i x_i = N\}$ and let

$$\mathcal{B}_N = \bigcup_{\lambda \in \mathbf{L}_N} \partial(\lambda + \mathbf{C}).$$

This countable union of $(d-2)$ -dimensional faces has induced $(d-1)$ -dimensional volume $|\mathcal{B}_N|_{\mathbf{A}_N} = 0$, and

$$\mathbf{A}_N \setminus \mathcal{B}_N = \bigsqcup_{\lambda \in \mathbf{L}_N} ((\lambda + \mathbf{C}) \setminus \mathcal{B}_N).$$

Thus every point outside $\mathcal{B}_N$ lies in exactly one translated cell.

More explicitly, let $T : \mathbb{R}^{d-1} \rightarrow \mathbf{H}$ be $Tt = \sum_{i < d} t_i b_i$. By definition, the Gram matrix of these coordinate vectors is $G = T^\top T$, whose entries are

$$G_{ij} = \left\langle \frac{\partial T}{\partial t_i}, \frac{\partial T}{\partial t_j} \right\rangle = \langle b_i, b_j \rangle = \langle e_i - e_d, e_j - e_d \rangle = \delta_{ij} + 1.$$

Thus G has diagonal entries 2 and off-diagonal entries 1. Equivalently,

$$G = I_{d-1} + \mathbf{1}\mathbf{1}^\top, \quad \mathbf{1} = (1, \dots, 1)^\top \in \mathbb{R}^{d-1}.$$

Its eigenvalue along $\mathbf{1}$ is d , while its eigenvalue on $\mathbf{1}^\perp$ is 1. Hence $\det G = d \cdot 1^{d-2} = d$.

To define the volume notation, choose an orthonormal basis $q_1, \dots, q_{d-1}$ of $\mathbf{H}$ and let $Q : \mathbb{R}^{d-1} \rightarrow \mathbf{H}$ be $Qs = \sum_{k < d} s_k q_k$. For a measurable $E \subset \mathbf{H}$, its induced Euclidean volume is

$$|E|_{\mathbf{H}} := |Q^{-1}(E)|_{\mathbb{R}^{d-1}} = \int_{\mathbb{R}^{d-1}} \mathbf{1}_E(Qs) \, ds_1 \cdots ds_{d-1}. \quad (2)$$

Here $\mathbf{1}_E(y) = 1$ for $y \in E$ and $\mathbf{1}_E(y) = 0$ for $y \notin E$. This does not depend on the chosen orthonormal basis, since two such bases are related by an orthogonal matrix, whose absolute determinant is 1. Equation (2) has no Jacobian factor because Q is an isometry. In contrast, the columns $b_i = e_i - e_d$ of T are not orthonormal. Since $Q^{-1} = Q^\top$ on $\mathbf{H}$, for every measurable $A \subset \mathbb{R}^{d-1}$,

$$|T(A)|_{\mathbf{H}} = |Q^{-1}T(A)|_{\mathbb{R}^{d-1}} = |\det(Q^{-1}T)| |A|_{\mathbb{R}^{d-1}}.$$

Moreover, $QQ^\top$ is the orthogonal projection onto $\mathbf{H}$ and $T(\mathbb{R}^{d-1}) = \mathbf{H}$, so

$$(Q^{-1}T)^\top(Q^{-1}T) = T^\top QQ^\top T = T^\top T = G.$$

Hence $|\det(Q^{-1}T)| = \sqrt{\det G}$, and therefore

$$|T(A)|_{\mathbf{H}} = \sqrt{\det G} |A|_{\mathbb{R}^{d-1}} = \sqrt{d} \int_A dt_1 \cdots dt_{d-1}. \quad (3)$$

Taking $A = [-1/2, 1/2)^{d-1}$, whose coordinate volume is 1, gives $|\mathbf{C}|_{\mathbf{H}} = \sqrt{d}$; below $|\mathbf{C}|$ denotes this induced volume.

2.1 Randomized rounding as cell membership

Let U_i be independent and uniform on $[-1/2, 1/2)$ and set

$$U = \sum_{i=1}^{d-1} U_i(e_i - e_d).$$

By (3), U is uniform with respect to induced volume in the concrete sense that

$$\Pr(U \in A) = \frac{|\mathbf{A}|_{\mathbf{H}}}{|\mathbf{C}|_{\mathbf{H}}} \quad \text{for every measurable } A \subset \mathbf{C}.$$

Given $\mu \in \mathbf{Y}_n$, define

$$\tilde{\nu}_i = \left\lfloor \gamma_n(\mu_i + U_i) + \frac{1}{2} \right\rfloor \quad (i < d), \quad \tilde{\nu}_d = m_n - \sum_{i < d} \tilde{\nu}_i.$$

Outside a null set of cell boundaries, this is equivalent to

$$\gamma_n(\mu + U) \in \tilde{\nu} + \mathbf{C}. \quad (4)$$

Indeed, for $i < d$, membership in the corresponding coordinate interval says that $\tilde{\nu}_i$ is the nearest integer to $\gamma_n(\mu_i + U_i)$. The last coordinate follows automatically because both vectors in (4) have coordinate sum m_n .

Let $D_n(\tilde{\nu} \mid \mu)$ be the distribution of the raw rounded label. Since U is uniform on $\mathbf{C}$,

$$D_n(\tilde{\nu} \mid \mu) = \frac{1}{|\mathbf{C}|} |\{u \in \mathbf{C} : \gamma_n(\mu + u) \in \tilde{\nu} + \mathbf{C}\}|. \quad (5)$$

Under the change of variables $x = \gamma_n(\mu + u)$, volume in the $(d-1)$ -dimensional space $\mathbf{H}$ is multiplied by γ_n^{d-1} . The image of the set in (5) is exactly

$$\gamma_n(\mu + \mathbf{C}) \cap (\tilde{\nu} + \mathbf{C}).$$

Therefore

$$D_n(\tilde{\nu} \mid \mu) = \frac{|\gamma_n(\mu + \mathbf{C}) \cap (\tilde{\nu} + \mathbf{C})|}{\gamma_n^{d-1} |\mathbf{C}|}. \quad (6)$$

Thus $D_n(\tilde{\nu} \mid \mu)$ is the fraction of the uniformly dilated input cell lying in the target cell indexed by $\tilde{\nu}$. The denominator is the volume of $\gamma_n(\mu + \mathbf{C})$.

3 Embedding lattice laws as densities

For $p \in \Delta_d^\circ$ and $\lambda \in \mathbf{L}_N$, define the rescaled cell

$$\mathbf{C}_{N,p}(\lambda) = \frac{\lambda - Np + \mathbf{C}}{\sqrt{N}} \subset \mathbf{H}.$$

Translation by $\lambda - Np$ does not change volume, while multiplication by $N^{-1/2}$ in the $(d-1)$ -dimensional space $\mathbf{H}$ multiplies volume by $(N^{-1/2})^{d-1}$. Hence

$$|\mathbf{C}_{N,p}(\lambda)| = (N^{-1/2})^{d-1} |\mathbf{C}| = \frac{|\mathbf{C}|}{N^{(d-1)/2}}.$$

These cells tile $\mathbf{H}$ up to null boundaries. Let $R : \mathbf{L}_N \rightarrow \mathbb{R}$ satisfy $\sum_{\lambda \in \mathbf{L}_N} |R(\lambda)| < \infty$; this is a finite signed measure on the discrete set $\mathbf{L}_N$. If additionally $R(\lambda) \geq 0$ and $\sum_{\lambda} R(\lambda) = 1$, then R is a probability law. Define

$$(\mathcal{E}_{N,p}R)(x) = \frac{N^{(d-1)/2}}{|\mathbf{C}|} R(\lambda), \quad x \in \mathbf{C}_{N,p}(\lambda). \quad (7)$$

The factor in (7) is the reciprocal cell volume. Hence

$$\int_{\mathbf{C}_{N,p}(\lambda)} (\mathcal{E}_{N,p}R)(x) dx = R(\lambda).$$

It follows cell by cell that

$$\|\mathcal{E}_{N,p}R - \mathcal{E}_{N,p}S\|_1 = \|R - S\|_{\ell^1}, \quad (8)$$

$$\int_{\mathbf{H}} \sqrt{(\mathcal{E}_{N,p}R)(x)(\mathcal{E}_{N,p}S)(x)} dx = \sum_{\lambda \in \mathbf{L}_N} \sqrt{R(\lambda)S(\lambda)}. \quad (9)$$

Thus the embedding preserves both total variation in the ℓ^1 convention and Hellinger affinity.

The limiting covariance on $\mathbf{H}$ is

$$\Sigma_p = \text{diag}(p) - pp^\top.$$

It has kernel spanned by $(1, \dots, 1)$ and is positive definite after restriction to $\mathbf{H}$. For $x \in \mathbf{H}$,

$$\langle x, \Sigma_p^{-1}x \rangle = \sum_i \frac{x_i^2}{p_i}, \quad \det_{\mathbf{H}} \Sigma_p = d \prod_i p_i. \quad (10)$$

Indeed, for $x \in \mathbf{H}$, the vector

$$z = \text{diag}(p)^{-1}x - \frac{1}{d} \left(\sum_i \frac{x_i}{p_i} \right) \mathbf{1}$$

satisfies $\sum_i z_i = \sum_i x_i/p_i - \sum_i x_i/p_i = 0$, so $z \in \mathbf{H}$. Since $\Sigma_p \mathbf{1} = 0$ and $\sum_i x_i = 0$,

$$\Sigma_p z = x - p \sum_i x_i = x, \quad \langle x, z \rangle = \sum_i \frac{x_i^2}{p_i} - \frac{1}{d} \left(\sum_i \frac{x_i}{p_i} \right) \sum_i x_i = \sum_i \frac{x_i^2}{p_i},$$

which proves the first identity. For the second, let $\Sigma_p^{(k)}$ be the matrix obtained by deleting row and column k . The matrix determinant lemma gives

$$\det \Sigma_p^{(k)} = \left(\prod_{i \neq k} p_i \right) \left(1 - \sum_{i \neq k} p_i \right) = \prod_i p_i.$$

Because the normalized kernel vector is $d^{-1/2}\mathbf{1}$, orthogonal diagonalization gives

$$\text{adj}(\Sigma_p) = \det_{\mathbb{H}}(\Sigma_p) \frac{\mathbf{1}\mathbf{1}^\top}{d}, \quad \det \Sigma_p^{(k)} = \frac{1}{d} \det_{\mathbb{H}} \Sigma_p.$$

Consequently, $\det_{\mathbb{H}} \Sigma_p = d \prod_i p_i$.

4 Lemma B.1: the uniform Young local limit

Lemma B.1 (Uniform Young local limit, rephrased). *For every compact $K \subseteq \Delta_d^\circ$,*

$$\sup_{p \in K} \|\mathcal{E}_{N,p} P_{N,p} - \varphi_{0,\Sigma_p}\|_1 \longrightarrow 0. \quad (11)$$

This is stronger than weak convergence. It says that after spreading each lattice mass uniformly over its cell, the entire density converges in L^1 , uniformly over p in a compact subset of the strictly ordered simplex.

Step 1: compare the Young law with a multinomial law

Let $X_1, \dots, X_N$ be independent random variables with $\Pr(X_k = i) = p_i$, and let $\Lambda_i = |\{k : X_k = i\}|$. Thus Λ_i counts outcomes of type i and $\sum_i \Lambda_i = N$. The law $M_{N,p}$ of $\Lambda = (\Lambda_1, \dots, \Lambda_d)$ is

$$M_{N,p}(\lambda) = \begin{cases} \frac{N!}{\prod_i \lambda_i!} \prod_i p_i^{\lambda_i}, & \lambda \in \mathbb{Z}_{\geq 0}^d, \sum_i \lambda_i = N, \\ 0, & \lambda \in \mathbb{L}_N \setminus \mathbb{Z}_{\geq 0}^d. \end{cases}$$

Define the central window

$$W_{N,p} = \left\{ \lambda \in \mathbb{L}_N : \|\lambda - Np\|_\infty \leq N^{2/3} \right\}.$$

Compactness of K gives uniform lower bounds on every p_i and every gap $p_i - p_{i+1}$. Consequently, every nonnegative composition in this window is a partition for all large N .

For a partition λ , the hook-length and Weyl character formulas give

$$\dim \mathcal{S}_\lambda = N! \frac{\prod_{i < j} (\lambda_i - \lambda_j + j - i)}{\prod_i (\lambda_i + d - i)!},$$

$$s_\lambda(p) = \frac{\det[p_i^{\lambda_j + d - j}]_{i,j}}{\prod_{i < j} (p_i - p_j)}.$$

On $W_{N,p}$, both the exponents $\lambda_j + d - j$ and the numbers $\log p_i$ are uniformly strictly decreasing. The rearrangement inequality therefore makes the identity-permutation term in the determinant strictly larger than every other term. The gaps are of order N , so the ratio of each nonidentity term to the identity term is $O(e^{-cN})$, uniformly for $p \in K$. Hence

$$s_\lambda(p) = \frac{\prod_i p_i^{\lambda_i + d - i}}{\prod_{i < j} (p_i - p_j)} (1 + O(e^{-cN})).$$

Dividing $P_{N,p}(\lambda)$ by $M_{N,p}(\lambda)$ now gives

$$\begin{aligned} \frac{P_{N,p}(\lambda)}{M_{N,p}(\lambda)} &= (1 + O(e^{-cN})) \prod_{i < j} \frac{\lambda_i - \lambda_j + j - i}{N(p_i - p_j)} \\ &\quad \times \prod_i \frac{\lambda_i! (Np_i)^{d-i}}{(\lambda_i + d - i)!}. \end{aligned} \quad (12)$$

Every factor in (12) is $1 + O_K(N^{-1/3})$ on the central window. For example,

$$\frac{\lambda_i!(Np_i)^{d-i}}{(\lambda_i + d - i)!} = \prod_{a=1}^{d-i} \frac{Np_i}{\lambda_i + a} = 1 + O_K(N^{-1/3}).$$

Therefore $P_{N,p}/M_{N,p} \rightarrow 1$ uniformly on $W_{N,p}$.

Outside the window, the uniform Young concentration estimate controls $P_{N,p}$ and Hoeffding's inequality controls $M_{N,p}$. Combining the central comparison with both tail estimates yields

$$\sup_{p \in K} \|P_{N,p} - M_{N,p}\|_{\ell^1} \rightarrow 0. \quad (13)$$

Step 2: prove a cellwise local limit for the multinomial law

Choose

$$\frac{1}{2} < \beta_0 < \frac{2}{3}, \quad x_\lambda = \frac{\lambda - Np}{\sqrt{N}}.$$

Uniform Stirling expansion and a third-order Taylor expansion give, whenever $\|\lambda - Np\|_\infty \leq N^{\beta_0}$,

$$\begin{aligned} \log M_{N,p}(\lambda) = & -\frac{d-1}{2} \log(2\pi N) - \frac{1}{2} \sum_i \log p_i - \frac{1}{2} \sum_i \frac{x_{\lambda,i}^2}{p_i} \\ & + O_K(N^{3\beta_0-2} + N^{\beta_0-1} + N^{-1}). \end{aligned} \quad (14)$$

The restriction $\beta_0 < 2/3$ makes the remainder tend to zero. Using (10), $|\mathbf{C}| = \sqrt{d}$, and $\det_{\mathbf{H}} \Sigma_p = d \prod_i p_i$, exponentiating (14) gives

$$\frac{N^{(d-1)/2}}{|\mathbf{C}|} M_{N,p}(\lambda) = \varphi_{0,\Sigma_p}(x_\lambda)(1 + o(1)) \quad (15)$$

uniformly on this smaller window.

Equation (15) compares the embedded constant on a cell with the Gaussian at the cell center. Three observations upgrade it to L^1 convergence:

1. Hoeffding's inequality makes the multinomial mass outside the window $O(e^{-c_K N^{2\beta_0-1}})$.
2. Uniform eigenvalue bounds for Σ_p give the same exponential bound for the Gaussian mass outside the corresponding region.
3. A rescaled cell has diameter $O(N^{-1/2})$. On the window, $\|\nabla \log \varphi_{0,\Sigma_p}(x)\| = O_K(1 + \|x\|)$, so the relative oscillation of the Gaussian across a cell is $\exp(O_K(N^{-1/2} + N^{\beta_0-1})) - 1 = o(1)$.

Thus

$$\sup_{p \in K} \|\mathcal{E}_{N,p} M_{N,p} - \varphi_{0,\Sigma_p}\|_1 \rightarrow 0. \quad (16)$$

Finally, the isometry (8), together with (13) and (16), proves (11).

5 Lemma B.2: randomized rounding becomes Gaussian dilation

Let $\tilde{Q}_{m_n,p} = P_{n,p}D_n$ be the raw output law: first draw $\mu \sim P_{n,p}$ and then $\tilde{\nu} \sim D_n(\cdot \mid \mu)$, before sending invalid outcomes to the fallback label.

Lemma B.2 (Continuous limit of randomized dilation, rephrased). *For every compact $K \subseteq \Delta_d^\circ$,*

$$\sup_{p \in K} \|\mathcal{E}_{m_n,p} \tilde{Q}_{m_n,p} - \varphi_{0,\gamma\Sigma_p}\|_1 \longrightarrow 0, \quad (17)$$

$$\sup_{p \in K} \left| B(\tilde{Q}_{m_n,p}, P_{m_n,p}) - B_{\text{cl}}(\gamma, d) \right| \longrightarrow 0. \quad (18)$$

Step 1: identify the exact continuous operation

For a density f on $\mathbb{H}$ and $a > 0$, define

$$(\text{dil}_a f)(y) = a^{-(d-1)/2} f(y/\sqrt{a}).$$

This is the density of $\sqrt{a}X$ when X has density f , and it is an L^1 isometry. Define cell averaging by

$$(\text{Av}_{m_n,p} f)(y) = \frac{1}{|\mathbb{C}_{m_n,p}(\nu)|} \int_{\mathbb{C}_{m_n,p}(\nu)} f(z) \, dz = \frac{m_n^{(d-1)/2}}{|\mathbb{C}|} \int_{\mathbb{C}_{m_n,p}(\nu)} f(z) \, dz, \quad y \in \mathbb{C}_{m_n,p}(\nu).$$

Thus $\text{Av}_{m_n,p} f$ is constant on every target cell. This operator is an L^1 contraction.

The central-limit coordinates of a point $\mu + u$ in an input cell obey

$$\frac{\gamma_n(\mu + u) - m_n p}{\sqrt{m_n}} = \sqrt{\gamma_n} \frac{\mu + u - np}{\sqrt{n}}. \quad (19)$$

The factor is $\sqrt{\gamma_n}$ rather than γ_n because the input and target central-limit scalings use $\sqrt{n}$ and $\sqrt{m_n}$. Equation (6) then says exactly which target cell contains the dilated point. Set $g_n = \text{dil}_{\gamma_n}(\mathcal{E}_{n,p} P_{n,p})$. On the dilated input cell $\sqrt{\gamma_n} \mathbb{C}_{n,p}(\mu)$, with $\mu \in \mathbb{Y}_n$,

$$g_n(y) = \gamma_n^{-(d-1)/2} (\mathcal{E}_{n,p} P_{n,p})(y/\sqrt{\gamma_n}) = \frac{n^{(d-1)/2}}{\gamma_n^{(d-1)/2} |\mathbb{C}|} P_{n,p}(\mu).$$

Equation (19) also gives

$$\sqrt{\gamma_n} \mathbb{C}_{n,p}(\mu) = \frac{\gamma_n(\mu + \mathbb{C}) - m_n p}{\sqrt{m_n}},$$

and therefore

$$|\mathbb{C}_{m_n,p}(\nu) \cap \sqrt{\gamma_n} \mathbb{C}_{n,p}(\mu)| = \frac{|(\nu + \mathbb{C}) \cap \gamma_n(\mu + \mathbb{C})|}{m_n^{(d-1)/2}}.$$

Multiplying the preceding constant density by this intersection volume and using $m_n = \gamma_n n$ and (6) gives

$$\begin{aligned} & \int_{\mathbb{C}_{m_n,p}(\nu) \cap \sqrt{\gamma_n} \mathbb{C}_{n,p}(\mu)} g_n(y) \, dy \\ &= P_{n,p}(\mu) \frac{|(\nu + \mathbb{C}) \cap \gamma_n(\mu + \mathbb{C})|}{\gamma_n^{d-1} |\mathbb{C}|} = P_{n,p}(\mu) D_n(\nu \mid \mu). \end{aligned}$$

Summing over the input cells, for every $\nu \in \mathcal{L}_{m_n}$,

$$\int_{\mathcal{C}_{m_n,p}(\nu)} g_n(y) \, dy = \sum_{\mu \in \mathcal{Y}_n} P_{n,p}(\mu) D_n(\nu \mid \mu) = \tilde{Q}_{m_n,p}(\nu).$$

Hence, for $y \in \mathcal{C}_{m_n,p}(\nu)$,

$$(\text{Av}_{m_n,p} g_n)(y) = \frac{\tilde{Q}_{m_n,p}(\nu)}{|\mathcal{C}_{m_n,p}(\nu)|} = \frac{m_n^{(d-1)/2}}{|\mathcal{C}|} \tilde{Q}_{m_n,p}(\nu) = (\mathcal{E}_{m_n,p} \tilde{Q}_{m_n,p})(y).$$

Thus

$$\mathcal{E}_{m_n,p} \tilde{Q}_{m_n,p} = \text{Av}_{m_n,p} \text{dil}_{\gamma_n}(\mathcal{E}_{n,p} P_{n,p}). \quad (20)$$

This is an identity, not merely an asymptotic approximation. Randomizing within the input cell is what makes the discrete kernel intertwine exactly with continuous dilation followed by target-cell averaging.

Step 2: pass to the Gaussian limit

Lemma B.1 and the L^1 isometry of dilation give

$$\text{dil}_{\gamma_n}(\mathcal{E}_{n,p} P_{n,p}) - \text{dil}_{\gamma_n} \varphi_{0,\Sigma_p} \longrightarrow 0$$

uniformly in L^1 . Moreover,

$$\text{dil}_{\gamma_n} \varphi_{0,\Sigma_p} = \varphi_{0,\gamma_n \Sigma_p} \longrightarrow \varphi_{0,\gamma \Sigma_p}$$

uniformly for $p \in K$. Finally, the target-cell diameter is $O(m_n^{-1/2})$. The compact family of Gaussians $\varphi_{0,a\Sigma_p}$, with $p \in K$ and a near γ , is uniformly translation-continuous in L^1 , so averaging it over the vanishing cells changes it by $o(1)$. Applying these three facts to (20) proves (17).

Step 3: take the affinity limit

Lemma B.1 at sample size m_n also gives

$$\mathcal{E}_{m_n,p} P_{m_n,p} \longrightarrow \varphi_{0,\Sigma_p}$$

uniformly in L^1 . For probability densities,

$$\|\sqrt{f} - \sqrt{g}\|_2^2 \leq \|f - g\|_1,$$

so Hellinger affinity is continuous under L^1 convergence. The affinity is preserved exactly by the cell embedding in (9). Therefore

$$\begin{aligned} \text{B}(\tilde{Q}_{m_n,p}, P_{m_n,p}) &\longrightarrow \int_{\mathcal{H}} \sqrt{\varphi_{0,\gamma \Sigma_p}(x) \varphi_{0,\Sigma_p}(x)} \, dx \\ &= \left(\frac{2\sqrt{\gamma}}{1+\gamma} \right)^{(d-1)/2} = B_{\text{cl}}(\gamma, d). \end{aligned}$$

The covariance Σ_p cancels in the centered Gaussian affinity, which explains why the classical factor depends only on γ and d .

6 From the raw kernel to the valid Young-label kernel

The raw output $\tilde{\nu}$ always lies in $\mathbb{L}_{m_n}$, but it need not be a partition, and $\tilde{\nu} - \mu$ need not be a partition of $m_n - n$. The actual kernel K_n^{rnd} keeps a raw outcome only when both conditions hold and the outcome is not the fallback label

$$\nu_n^{\text{fb}} = (m_n, 0, \dots, 0).$$

All rejected mass is moved to this fallback label.

Randomized Young-label dilation (rephrased). *There is $c_{\text{rnd}} < \infty$ such that, uniformly for p in a compact $K \subseteq \Delta_d^\circ$, every sufficiently large n , μ in the typical set*

$$\mathbb{T}_n(p) = \left\{ \mu \in \mathbb{Y}_n : \|\mu/n - p\|_\infty \leq n^{-1/3} \right\},$$

and every ν with $K_n^{\text{rnd}}(\nu \mid \mu) > 0$ satisfy

$$\nu - \mu \vdash m_n - n, \quad \|\nu - \gamma_n \mu\|_\infty \leq c_{\text{rnd}}.$$

If

$$Q_{m_n, p}(\nu) = \sum_{\mu} P_{n, p}(\mu) K_n^{\text{rnd}}(\nu \mid \mu),$$

then

$$\sup_{p \in K} |B(Q_{m_n, p}, P_{m_n, p}) - B_{\text{cl}}(\gamma, d)| \longrightarrow 0.$$

Step 1: the rounding error is uniformly bounded

For $i < d$, nearest-integer rounding gives

$$|\tilde{\nu}_i - \gamma_n \mu_i| \leq \frac{\gamma_n + 1}{2}.$$

The last-coordinate errors sum to the negative of the first $d - 1$ errors, so

$$|\tilde{\nu}_d - \gamma_n \mu_d| \leq \frac{(d - 1)(\gamma_n + 1)}{2}.$$

Since $\gamma_n \rightarrow \gamma$, one may take

$$c_{\text{rnd}} = \frac{d - 1}{2} \left(1 + \sup_n \gamma_n \right) < \infty. \quad (21)$$

Step 2: typical labels need no fallback correction

Compactness of K inside the strictly ordered simplex gives $c_0 > 0$ such that, for all large n and every $\mu \in \mathbb{T}_n(p)$,

$$\mu_d \geq c_0 n, \quad \mu_i - \mu_{i+1} \geq c_0 n \quad (i < d).$$

Set $\delta = \tilde{\nu} - \mu$. It is an integer vector with coordinate sum $m_n - n$. The rounding bound gives

$$\begin{aligned} \delta_d &\geq (\gamma_n - 1)\mu_d - c_{\text{rnd}}, \\ \delta_i - \delta_{i+1} &\geq (\gamma_n - 1)(\mu_i - \mu_{i+1}) - 2c_{\text{rnd}}. \end{aligned}$$

Because $\gamma_n - 1$ is bounded away from zero for large n , both right-hand sides are positive. Hence

$$\delta_1 \geq \dots \geq \delta_d > 0,$$

so $\delta \vdash m_n - n$. Since μ and δ are partitions, $\tilde{\nu} = \mu + \delta$ is also a partition. Moreover, $\tilde{\nu}_d > \mu_d > 0$, whereas the fallback label has last coordinate zero. Thus $D_n(\cdot \mid \mu)$ is supported entirely on accepted labels and assigns no mass to the fallback label. For such typical μ ,

$$K_n^{\text{rnd}}(\nu \mid \mu) = D_n(\nu \mid \mu).$$

Step 3: replacement changes only negligible mass

Let $\varepsilon_{n,p}$ be the probability that the raw outcome is replaced when $\mu \sim P_{n,p}$. Step 2 shows that replacement can occur only when $\mu \notin \mathsf{T}_n(p)$, so

$$\varepsilon_{n,p} \leq P_{n,p}(\mathsf{T}_n(p)^c) \longrightarrow 0$$

uniformly on K by uniform Young concentration.

Let $\tilde{Q}_{m_n,p}$ be the raw output law and $Q_{m_n,p}$ the law after replacement. Moving mass $\varepsilon_{n,p}$ from rejected labels to one fallback label changes the ℓ^1 distance by at most twice that mass:

$$\|Q_{m_n,p} - \tilde{Q}_{m_n,p}\|_{\ell^1} \leq 2\varepsilon_{n,p} = o(1).$$

The continuity of Hellinger affinity and Lemma B.2 now give

$$\mathsf{B}(Q_{m_n,p}, P_{m_n,p}) = \mathsf{B}(\tilde{Q}_{m_n,p}, P_{m_n,p}) + o(1) = B_{\text{cl}}(\gamma, d) + o(1),$$

uniformly for $p \in K$. This proves the proposition.

7 How the pieces fit together

The logical flow of Appendix B is

$$\begin{array}{c} P_{N,p} \text{ is uniformly close to the multinomial law} \\ \Downarrow \\ \mathcal{E}_{N,p} P_{N,p} \rightarrow \varphi_{0,\Sigma_p} \text{ in } L^1 \quad \text{Lemma B.1} \\ \Downarrow \\ \text{randomized rounding} = \text{dilation by } \sqrt{\gamma_n} + \text{cell averaging} \\ \Downarrow \\ \mathcal{E}_{m_n,p} \tilde{Q}_{m_n,p} \rightarrow \varphi_{0,\gamma\Sigma_p} \quad \text{Lemma B.2} \\ \Downarrow \\ \mathsf{B}(\tilde{Q}_{m_n,p}, P_{m_n,p}) \rightarrow B_{\text{cl}}(\gamma, d) \\ \Downarrow \\ \text{the fallback changes } o(1) \text{ mass} \\ \Downarrow \\ \mathsf{B}(Q_{m_n,p}, P_{m_n,p}) \rightarrow B_{\text{cl}}(\gamma, d). \end{array}$$

The randomized cell variable is essential: it turns discrete rounding into the exact intertwining identity (20). The typical-set argument is logically separate. It does not establish the Gaussian limit; it shows that the raw limiting construction is compatible with the Young branching rule except on an event of vanishing probability.

8 Points worth keeping straight

1. The raw kernel D_n is defined on the full affine lattice. The corrected kernel K_n^{rnd} is a Markov kernel on Young labels.
2. The power γ_n^{d-1} in (6) is a Jacobian in the $(d-1)$ -dimensional hyperplane $\mathbf{H}$.
3. Dilation by γ_n in the original labels becomes dilation by $\sqrt{\gamma_n}$ after central-limit normalization.
4. Lemma B.1 is an L^1 local limit, not merely convergence in distribution. This strength is what permits passage to Hellinger affinity.
5. Compactness of $K \subseteq \Delta_d^\circ$ supplies uniform lower bounds for the coordinates p_i , the spectral gaps $p_i - p_{i+1}$, and the eigenvalues of Σ_p on $\mathbf{H}$.
6. The fallback label is needed to define a valid kernel for every input label, but it is asymptotically invisible because typical labels never use it.